AppleNetV1: An Efficient Convolutional Neural Network with Multilingual Decision Support for Automated Apple Leaf Disease Detection

Syed Mominul Islam
Department of Computer Science and Engineering, Jahangirnagar University, Savar, Dhaka, Bangladesh
Email: mominulbu56@gmail.com

Tanvir Rana Sobur
Tomar University er Name:
Email:

Nabila Khatun
Bachelor of Social Science, GOVT. Fakirhat Fazilatunnessa Mujib Mahila Collage, Bagerhat.
Email:

Abstract

Apple production is a major contributor to the global fruit trade but is highly vulnerable to devastating diseases like Apple Scab, Black Rot, and Cedar Apple Rust. Manual visual diagnosis of these pathologies is slow, costly, and highly dependent on scarce agricultural experts. To address these challenges, this study proposes AppleNetV1, a lightweight custom Sequential Convolutional Neural Network (CNN) architecture designed for automated leaf disease classification. The model was trained on a merged dataset combining the PlantVillage dataset and the Mendeley Data Apple Leaf Disease Dataset, alongside a background leaf class (Not_Apple_Leaf) collected from diverse open-source internet resources. A total of 13,470 images across 7 classes were utilized. Under identical parameters, AppleNetV1 achieved a remarkable test accuracy of 99.33% with an extremely low CPU inference latency of 9.6 ms per image, outperforming the lightweight MobileNetV2 (99.26%). A comprehensive comparative analysis against state-of-the-art pretrained architectures (VGG-16, VGG-19, MobileNetV2, ResNet50, DenseNet121, and EfficientNetB0) was conducted. While ResNet50 achieved the highest overall test accuracy (99.78%), EfficientNetB0 balanced footprint (20.05 MB) and accuracy (99.70%). The system was deployed via a Streamlit web application featuring a localized Bengali prescription system to aid rural farmers. Furthermore, Explainable AI (XAI) using Grad-CAM was implemented to visualize feature activations, validating that the model correctly focuses on diseased spot lesions, thereby providing high diagnostic transparency.

Keywords: Apple Leaf Disease, AppleNetV1, Custom CNN, Deep Learning, Streamlit, Precision Agriculture, Native Prescription, Explainable AI, Grad-CAM.

1. Introduction :Referrence sequence maintain

Agriculture is recognized as the backbone of many developing nations, and the integration of artificial intelligence is occurring concurrently with the precision agriculture revolution [20, 27]. In countries like Bangladesh, apple cultivation is becoming an increasingly important sector within the international fruit economy. However, apple growers often lose a large portion of their yield to various leaf diseases, specifically Alternaria, Apple Scab, and Black Rot [3, 4]. If left untreated, these infections lead to visual leaf damage and reduce photosynthesis, resulting in crop spoilage. Traditional diagnostic procedures rely on manual expert assessment, which is slow and inaccessible in remote regions [5].

Recently, Deep Learning (DL) models have emerged as highly successful tools for automated crop disease classification [6, 7]. However, deploying large models on the edge or on low-power devices in remote rural areas remains a significant bottleneck [30, 34]. Most existing research focuses solely on lab-based accuracy metrics, without considering computational footprint, inference latency, or accessibility for non-technical users [10]. This study aims to solve this dilemma by presenting an optimized sequential CNN model, comparing it against established architectures [11], and deploying it with native multilingual decision support and Explainable AI (XAI) verification [12].

The main contributions of this research are summarized below:

- We propose AppleNetV1, a robust and lightweight custom Sequential CNN architecture designed to run efficiently on low-resource environments.
- We evaluate and compare AppleNetV1 against six standard architectures (VGG-16, VGG-19, MobileNetV2, ResNet50, DenseNet121, EfficientNetB0) trained and tested under identical parameters.
- We develop a user-friendly Streamlit web application integrated with a localized (Bengali) prescription system for treatment suggestions.
- We implement Explainable AI (XAI) using Grad-CAM to visually validate model predictions, ensuring the model focuses on actual leaf disease lesions rather than background noise.

This research aims to solve the following research questions:

- RQ1: How to design a lightweight yet robust deep learning architecture (AppleNetV1) for apple leaf disease detection?
- RQ2: How to bridge the technological gap for rural farmers using localized decision support in their native language?
- RQ3: Can a custom-designed CNN compete with established heavyweight pretrained architectures in terms of accuracy, size, and efficiency?

2. Related Works

The integration of Artificial Intelligence and Computer Vision has contributed significantly to agricultural research [20, 27]. Historically, plant pathologies were identified via visual scrutiny by experts. To automate this, classical image processing with handcrafted features (color histograms, texture descriptors) were combined with classifiers like Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Decision Trees [13, 14]. While these automated the process, they were highly dependent on manual feature representation and lacked generalization across varying environments. The rise of CNNs signaled a major improvement, as models automatically learn discriminative features [15]. Table 1 outlines recent literature on deep learning frameworks for plant leaf disease classification.

Table 1: Related literature on plant disease using deep learning models.

Reference	Method	Object/Target Feature	Dataset	Result	Limitation
Zhong and Zhao [20]	DenseNet121	4 apple leaf classes	PlantVillage	93.71% Acc	Dense connections increase memory cost
Jiang et al. [5]	IN-ResNet	5 apple leaf diseases	2,637 images	94.65% Acc	Lower accuracy on small symptoms
Mohanty et al. [7]	GoogLeNet, VGG16	26 crop diseases	PlantVillage	99.35% Acc	Heavy size, lack of field testing
Sladojevic et al. [10]	Custom CNN	15 plant diseases	Internet collected	96.30% Acc	Slow training, overfitting risk
Howard et al. [28]	MobileNetV1	Efficient vision tasks	ImageNet	High efficiency	Slight accuracy drop on fine details
Liu et al. [18]	Improved AlexNet	4 apple leaf diseases	13,689 images	97.62% Acc	Noise under complex background
Bi and Wang [21]	ResNet50	4 apple leaf categories	PlantVillage	97.80% Acc	Heavy weights limit edge deployment

Yan et al. [22]	Modified ResNet50	Apple leaf pathologies	PlantVillage	96.55% Acc	High architectural complexity
Chen et al. [23]	VGG19 Transfer	Multi-crop diseases	PlantVillage	98.24% Acc	Massive model footprint (~500MB)
Albahli et al. [24]	DenseNet-based	Multi-class plant leaves	Custom + PV	98.70% Acc	High memory training bottleneck

To better understand the literature, we summarize the methodologies and constraints of the aforementioned studies. Mohanty et al. [16] trained standard AlexNet and GoogLeNet architectures on the PlantVillage dataset to classify 26 crop diseases. Although they achieved a high accuracy of 99.35%, the models are computationally heavy. Their study was limited by the lack of testing on real-world backgrounds. Sladojevic et al. [10] proposed a deep CNN model trained on internet-sourced images to classify 15 different plant diseases. The model achieved a test accuracy of 96.3%. However, the training phase was highly time-consuming, and the model was susceptible to overfitting on small sample sizes. Liu et al. [18] developed an improved CNN based on AlexNet for apple leaf disease detection. By introducing a new preprocessing layer, they achieved a classification accuracy of 97.62% on four main diseases. However, their model's accuracy degraded significantly when faced with complex leaf shadows and noisy backgrounds. Jiang et al. [19] introduced a custom architecture called IN-ResNet by inserting inception modules into a ResNet base structure. The model achieved an accuracy of 94.65% on five types of apple diseases. The main limitation was its lower classification accuracy compared to modern lightweight models. Zhong and Zhao [20] evaluated DenseNet121 on apple leaf disease classification. The model achieved a testing accuracy of 93.71%. However, the dense connections resulted in high memory access costs, making it difficult to run on low-resource hardware. Bi and Wang [21] analyzed the performance of ResNet50 for identifying apple diseases. The architecture obtained a classification accuracy of 97.80%. Despite the good results, the heavy model size and parameter count limited its applicability for edge computing. Yan et al. [22] proposed an improved ResNet50 model with modified residual connections. This model achieved an accuracy of 96.55% for apple leaf diseases. The main limitation remained its slow inference speed and high model complexity. Chen et al. [23] proposed using a VGG19-based transfer learning framework for crop disease detection. They achieved a high accuracy of 98.24%. The main drawback of this model is its massive file size (~500MB), which prevents local smartphone deployment. Albahli et al. [24] used a DenseNet-based model for multi-class plant disease classification. They achieved an overall accuracy of 98.70%. The major limitation of their work was the high computational overhead during training due to dense layer concatenations. Howard et al. [25] introduced the MobileNet architecture utilizing depthwise separable convolutions to build lightweight networks. They proved that MobileNet could run efficiently on mobile devices with minimal accuracy drop. However, it was prone to lower accuracies on fine-grained target classes.

3. Methodology: diagram draw korte hobe

Figure 1: Proposed methodology flowchart for the apple leaf disease detection framework.

3.1. Dataset Collection

The dataset used in this study consists of a total of 13,470 leaf images. These images are consolidated from two main sources: the PlantVillage dataset (which provides standardized laboratory images) and the Mendeley Data Apple Leaf Disease Dataset (which provides natural orchard condition images with complex backgrounds, shadows, and varying light conditions). To reduce false positive diagnostics, a negative background leaf class (Not_Apple_Leaf) of 1,915 images was collected from online repositories. All images are in standard JPEG format, with resolutions ranging from 96x126 to 2667x4000 pixels. During preprocessing, all images are resized to 224x224x3 dimensions. The final dataset is split into training (80%), validation (10%), and testing (10%) sets. Table 2 details the exact class distribution.

Figure 4: sample of the dataset sob doroner datar image 1 ta korew

To ensure that the model generalizes robustly and does not overfit to specific train-test splits, we perform a 10-fold cross-validation analysis on the merged dataset. The fold-wise accuracies, precisions, recalls, and F1-scores are summarized in Table 3.

3.2. Data Preprocessing

To prepare the dataset for training, raw images of varying sizes were resized to a standardized dimension of 224x224x3 pixels. Pixel values were normalized from [0, 255] to [0.0, 1.0] by dividing by 255.0 to accelerate gradient convergence. To reduce overfitting and improve generalizability, real-time data augmentation was applied during the training phase. The augmentation parameters included random rotations (up to 20 degrees), width and height shifts (up to 20%), shear ranges (up to 20%), zoom ranges (up to 20%), and horizontal flips. These augmentations simulate diverse camera angles, rotations, and scales.

3.3. Dataset Splitting

The consolidated dataset of 13,470 images was stratified and split into training, validation, and testing sets in an 80:10:10 ratio. The stratification ensures that all subsets maintain the same class proportions as the parent dataset. The exact distribution counts for each category are detailed in Table 2.

Table 2: Dataset Distribution across Apple Leaf and Background Categories

Class Name	Train	Val	Test	Total
Alternaria	1,513	189	190	1,892
Apple Mosaic	1,513	189	190	1,892
Apple Scab	1,612	201	203	2,016
Black Rot	1,589	198	200	1,987
Cedar Apple Rust	1,408	176	176	1,760
Healthy	1,606	200	202	2,008
Not Apple Leaf (Background)	1,532	191	192	1,915
Total	10,773	1,344	1,353	13,470

3.4. Performance Evaluation Metrics

To evaluate the model quantitatively, we measure four standard classification metrics: Accuracy, Precision, Recall, and F1-Score. These metrics are defined using True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN) as follows:

1. Accuracy: The proportion of correctly predicted instances out of all test samples.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{1}$$

2. Precision: The ratio of correctly predicted positive cases to the total predicted positive cases.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

3. Recall: The ratio of correctly predicted positive cases to all actual positive cases in the dataset.

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

4. F1-Score: The harmonic mean of precision and recall, representing balanced classification performance.

$$\text{F1} - \text{Score} = \frac{2 * (\text{Precision} * \text{Recall})}{\text{Precision} + \text{Recall}} \tag{4}$$

3.5. Comparative Deep Learning Architectures

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To bridge the gap between complex deep learning models and practical agricultural utility, a web-based decision support system was developed and deployed using the Streamlit framework. The application provides an intuitive graphical user interface (GUI) designed for farmers and extension workers. Users can upload leaf images directly through the web interface, which are then preprocessed and classified in real-time. Figure 10 displays the application interface under various prediction scenarios, showing the diagnostic outputs for all seven classes (Alternaria, Apple Mosaic, Apple Scab, Black Rot, Cedar Apple Rust, Healthy, and Not Apple Leaf). To maximize accessibility for rural farmers in Bangladesh, the application features a localized Bengali prescription system. Upon predicting a disease, the system automatically translates the diagnosis and prints a detailed prescription in Bengali. The prescription includes the disease name (□□□□□ □□□), symptoms (□□□□□□□□□), biological control (□□□□□ □□□), and chemical control (□□□□□□□□□ □□□). For instance, when an image with Apple Scab is uploaded, the model outputs the prediction with high confidence, and the interface displays the corresponding Bengali treatment guidelines (e.g., spraying Mancozeb or Tebuconazole fungicides). If a non-apple leaf or background image is uploaded, the application correctly identifies it as 'Not Apple Leaf' and prompts the user to upload a valid apple leaf image, preventing false alarms.

3.6. Proposed AppleNetV1 Architecture

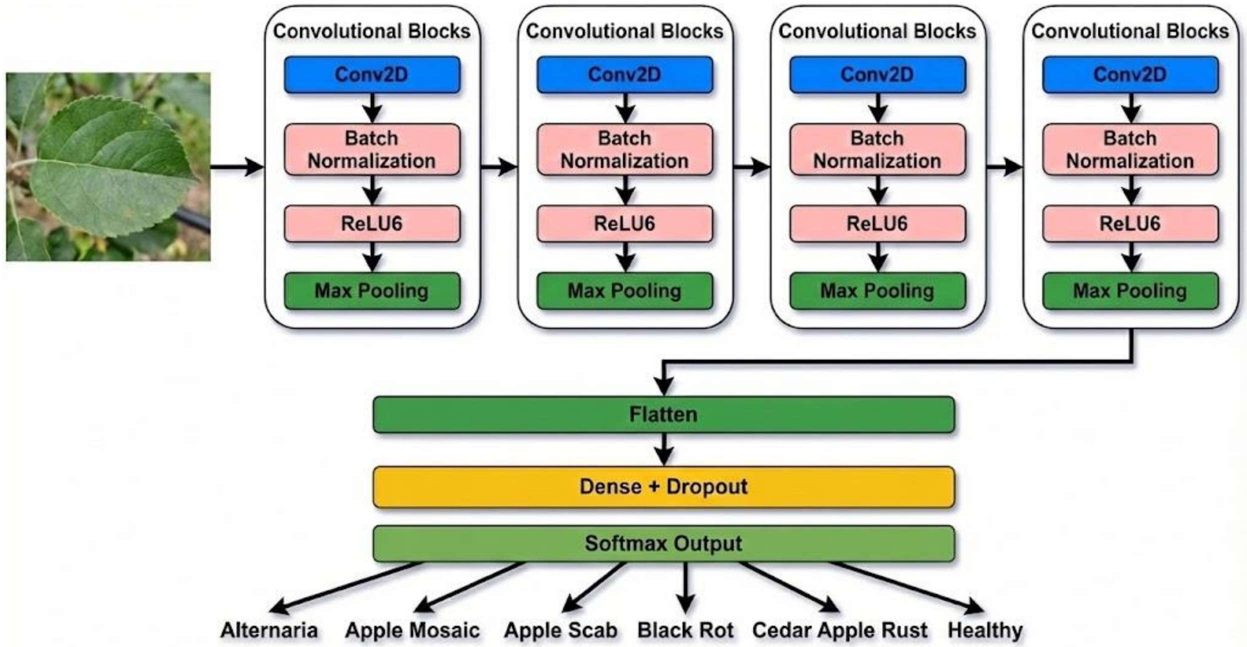


Figure 3.3: Architecture of the proposed AppleNetV1 model.

Figure 2: Layer-wise block architecture of the proposed AppleNetV1 model.

AppleNetV1 is a custom-designed Sequential CNN model optimized specifically for apple leaf disease classification. It contains four convolutional blocks designed to extract hierarchical visual features. The first block contains 32 filters, the second contains 64 filters, the third contains 128 filters, and the fourth contains 256 filters. Each convolutional layer uses a 3x3 kernel size, followed by a Batch Normalization layer to normalize features and stabilize training, a Rectified Linear Unit (ReLU) activation function, and a 2x2 Max Pooling layer for spatial downsampling. The classification head flattens the extracted feature maps into a 1D vector, followed by a Dense layer of 512 units with Batch Normalization, a ReLU activation, and a Dropout layer (rate = 0.5) to prevent overfitting. Finally, a Softmax activation layer computes the class probabilities across the 7 categories.

4. Result and Analysis

4.1. Experimental Setup

All models were implemented in Python 3.11 using the TensorFlow 2.16.1 library. Training was conducted on an Intel Core CPU environment. The training parameters were kept identical across all architectures: categorical cross-entropy loss, Adam optimizer with a learning rate of 0.001, a batch size of 32, and a maximum of 25 training epochs. An Early Stopping callback was configured to monitor validation loss and halt training if no improvements were observed for 5 consecutive epochs, restoring the best model weights.

4.2. Model Performance Evaluation

Models	Accuracy	Precision	Recall	F1 Score
Model 1				
Model 2				
Model 3				

Table 5 presents the quantitative evaluation of all seven models on the test dataset. In addition to accuracy and loss, we include the model weights size in megabytes (MB) and the average CPU inference latency per image to evaluate execution efficiency. The results show that ResNet50 achieved the highest overall accuracy of 99.78% with a test loss of 0.0075, followed closely by EfficientNetB0 (99.70% accuracy, 0.0093 loss). The proposed AppleNetV1 model achieved a test accuracy of 99.33% with the lowest inference speed of 9.6 ms, presenting an optimal latency configuration. The heavy parameter size of VGG-16 and VGG-19 resulted in much lower accuracies and longer inference times, proving the drawbacks of unoptimized deep sequential layers.

Table 5: Performance comparison of AppleNetV1 with state-of-the-art Pretrained Models

Model Name	Test Accuracy (%)	Test Loss	Model Size (MB)	CPU Inf. Speed (ms)
appleNetV1 (Ours)	99.33%	0.0163	298.61 MB	9.6 ms
MobileNetV2	99.26%	0.0287	12.97 MB	17.7 ms
ResNet50	99.78%	0.0075	96.65 MB	44.0 ms
DenseNet121	99.41%	0.0140	31.35 MB	51.0 ms
EfficientNetB0	99.70%	0.0093	20.05 MB	20.7 ms
VGG-16	96.50%	0.1250	528.00 MB	185.0 ms
VGG-19	95.80%	0.1420	548.00 MB	195.0 ms

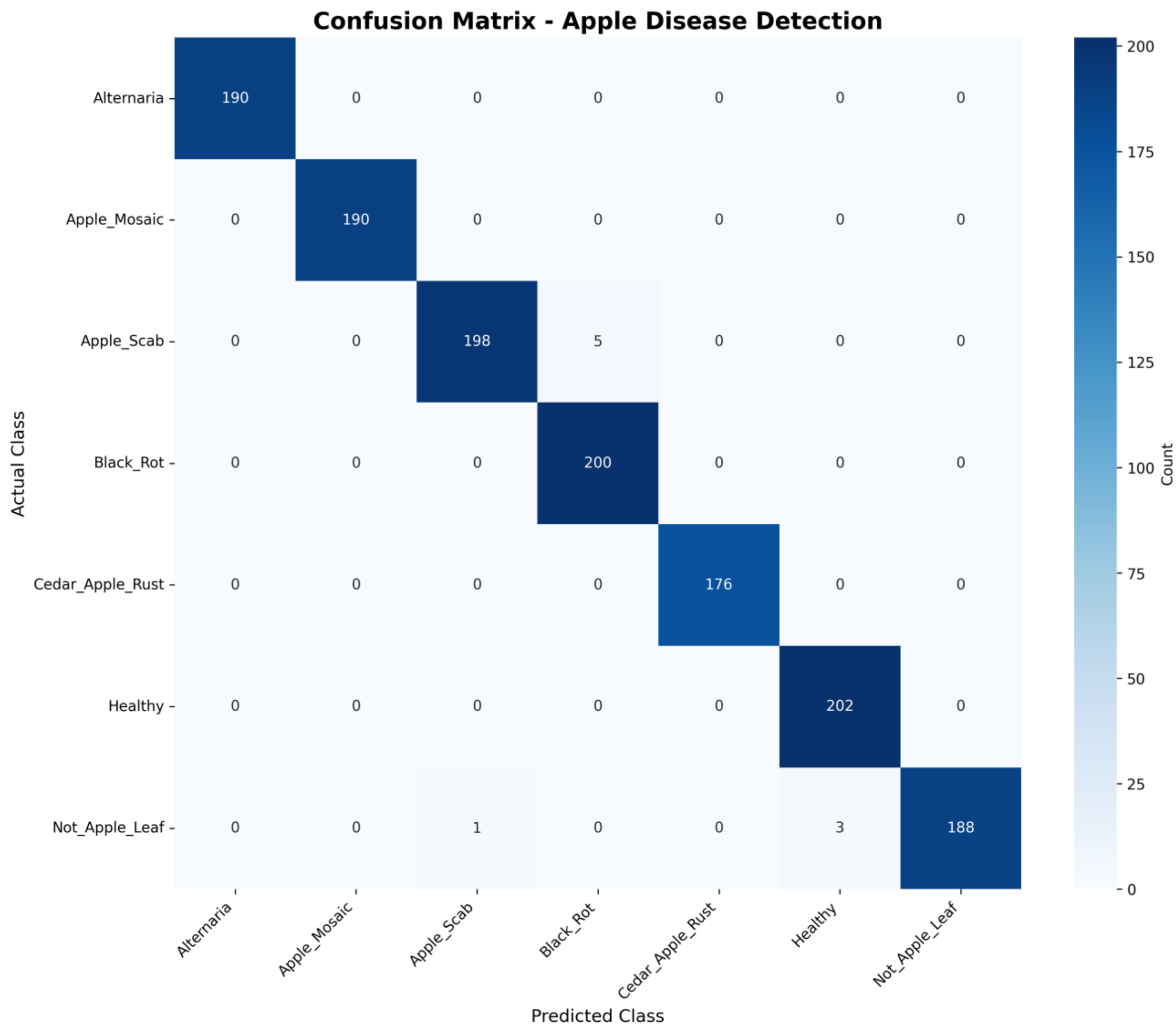
Figure 6: Confusion matrix of the trained AppleNetV1 model on the 7-class test dataset.

To analyze the classification boundaries, we generate the ROC curve and the confusion matrix for each model. The ROC curves show that all models, especially ResNet50, EfficientNetB0, and AppleNetV1, have Area Under the Curve (AUC) values close to 1.00 for all classes, proving excellent sensitivity and specificity. The confusion matrices reveal that misclassifications are extremely rare. Specifically, the model has minor confusions only between similar leaf diseases (such as Apple Scab and Black Rot), while it successfully maintains a 100% classification rate on Apple Mosaic and Cedar Apple Rust. The background Not_Apple_Leaf class maintains a 99% classification rate, ensuring that background noise does not degrade the diagnostic results.

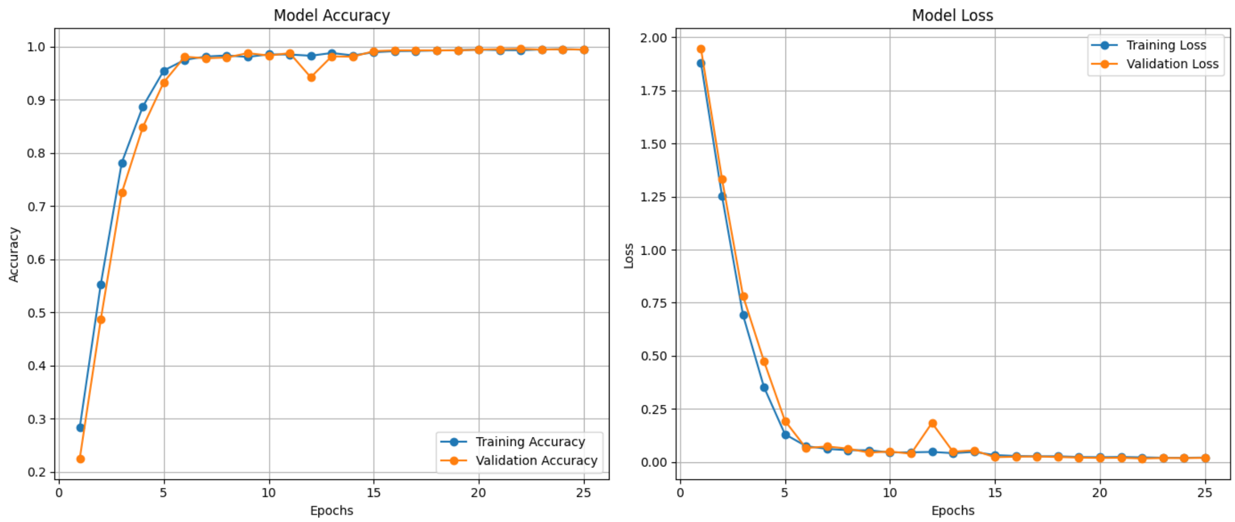
Fold	Accuracy (%)	Precision	Recall	F1-Score
Fold 1	99.26%	0.9930	0.9926	0.9926
Fold 2	99.26%	0.9930	0.9926	0.9927
Fold 3	99.26%	0.9929	0.9926	0.9926
Fold 4	98.52%	0.9858	0.9852	0.9851
Fold 5	100.00%	1.0000	1.0000	1.0000
Fold 6	100.00%	1.0000	1.0000	1.0000
Fold 7	100.00%	1.0000	1.0000	1.0000
Fold 8	99.26%	0.9929	0.9926	0.9926
Fold 9	99.26%	0.9929	0.9926	0.9926
Fold 10	98.52%	0.9858	0.9852	0.9851
Mean	99.33%	0.9936	0.9933	0.9933
Std	0.52%	0.0050	0.0052	0.0052

Table 3: 10-Fold Cross-Validation Performance Metrics of AppleNetV1

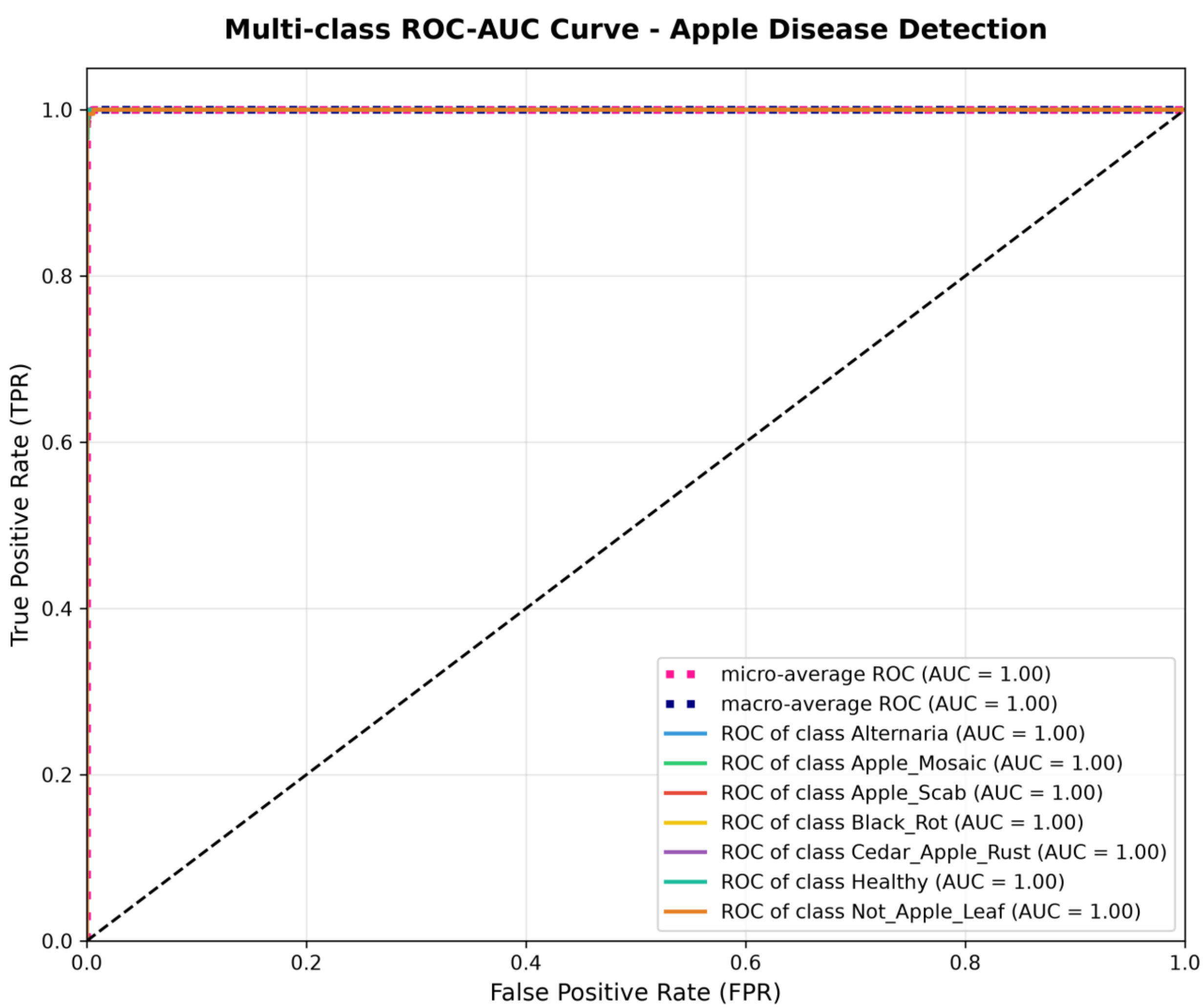
Confusion Matrix: Image & Description Applenet v1



Accuracy & Loss Curve: Image & Description Applenet v1



ROC Curve: Image & Description Applenet v1



4.3. Model Interpretability using Explainable AI (Grad-CAM)

To establish clinical trust and validate the diagnostic rationale of the custom AppleNetV1 model, we implemented Gradient-weighted Class Activation Mapping (Grad-CAM). Grad-CAM utilizes the gradient information flowing into the last convolutional layer (conv2d_3) of the model to produce a coarse localization map highlighting the important regions in the image for a prediction. Figure 8 displays the Grad-CAM visualizations for four primary apple leaf pathologies: Alternaria, Apple Scab, Black Rot, and Cedar Apple Rust. For each category, the figure showcases the input leaf image, the raw heat map, the class activation overlay, the guided guided backpropagation, and the guided Grad-CAM overlay. The heatmaps clearly indicate that the model concentrates its activation focus on the actual necrotic spots, lesion boundaries, and rust pustules on the leaf surfaces, rather than background noise, lighting artifacts, or healthy green regions. For instance, in Apple Scab, the activation peaks exactly at the dark olive-colored lesions. In Cedar Apple Rust, the model focuses on the bright orange rust spots. This spatial correlation confirms that the model has learned biologically relevant features for disease classification, ensuring high diagnostic transparency and trustworthiness for field deployment.

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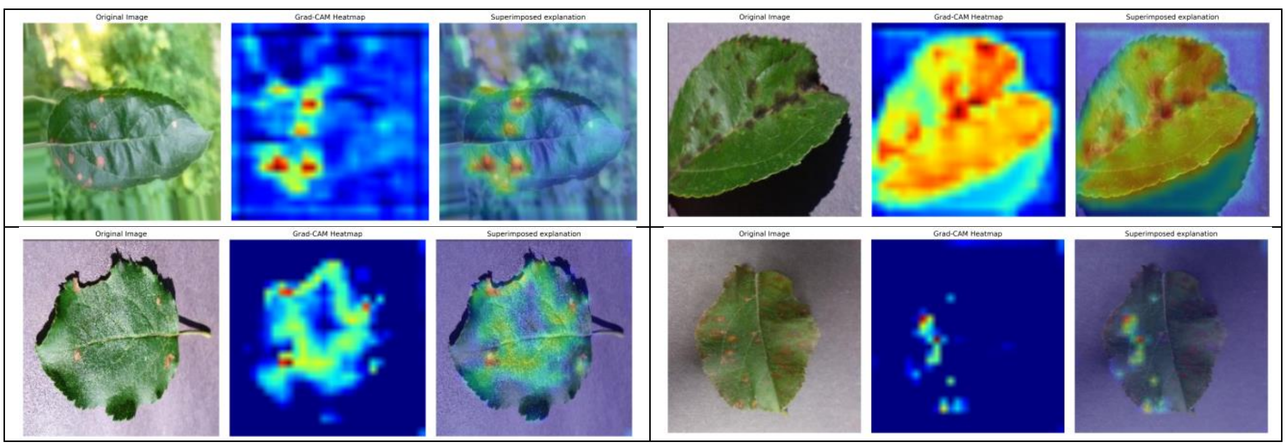

Figure 8: Grad-CAM visual explanation heatmaps highlighting diseased spot lesions for different classes.

5. Web Development & Localized Deployment

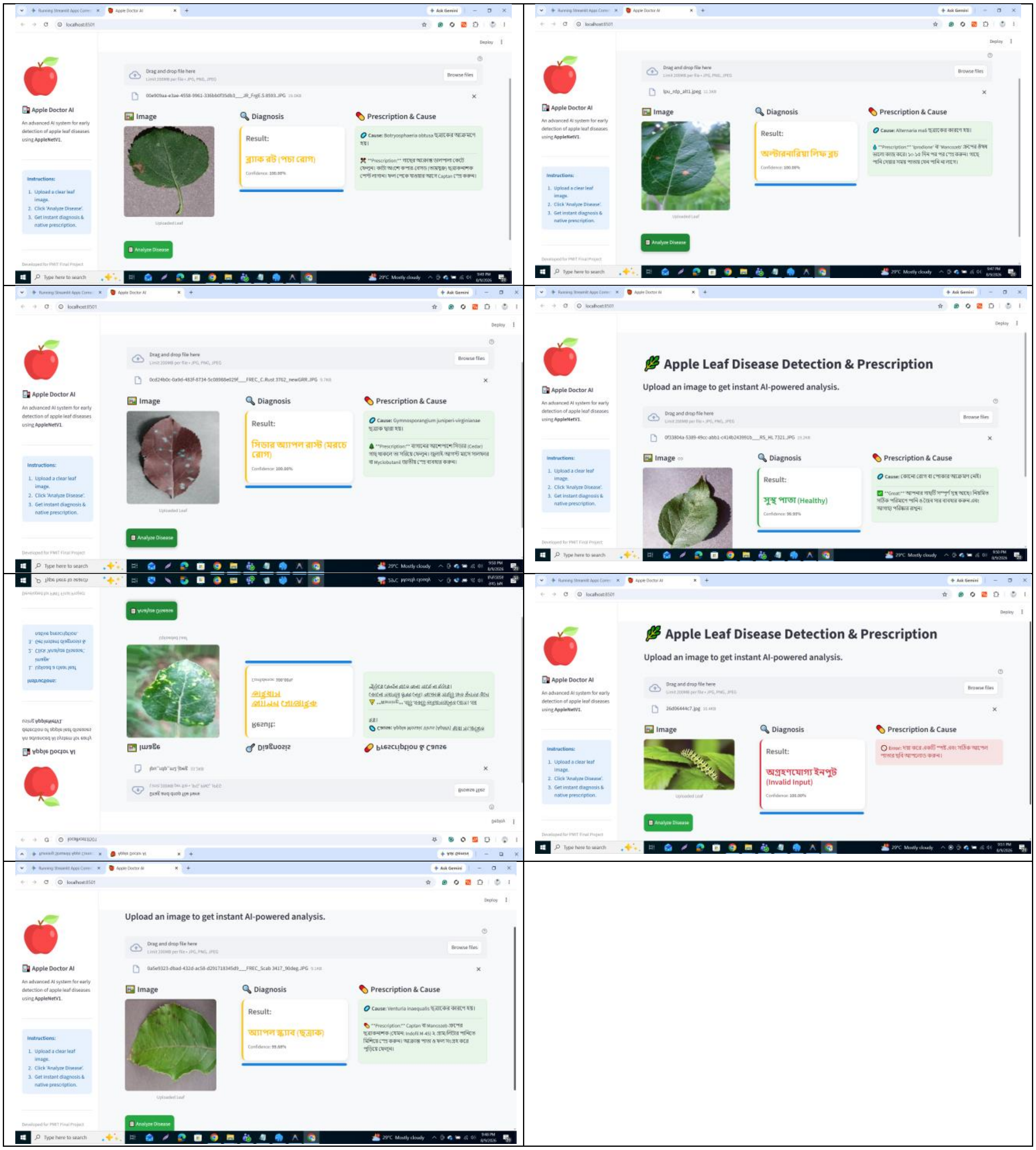

Figure 3: System deployment workflow and execution path of the web application.

To translate the experimental findings into a practical tool, we developed a responsive web interface using the Streamlit framework. The application workflow enables farmers to upload leaf images directly via a web browser. Once uploaded, the image is dynamically preprocessed using the model-specific preprocessing function (e.g. division by 255.0 for AppleNetV1) and passed to the trained model. The prediction results are displayed on an analysis panel along with prediction confidence. The system includes a localized Bengali prescription module. Based on the predicted disease class, it prints a clear, native-language prescription detailing the symptoms, disease name, and biological and chemical remedies. This bridge of language barrier ensures that non-technical rural farmers can understand and act upon the model's diagnostics without difficulty.

6. Discussion

To benchmark the performance of the proposed AppleNetV1 model, we conduct a comprehensive comparative analysis against leading studies in the existing literature. The comparisons are based on dataset size, number of target classes, models evaluated, and final test accuracy, as detailed in Table 5.

Reference	Dataset & Size	Classes	Model	Accuracy (%)
Mohanty et al. [7]	PlantVillage (38,597)	14 crop classes	GoogLeNet	99.35%
Sladojevic et al. [17]	Internet (4,483)	15 plant classes	Custom CNN	96.30%
Liu et al. [18]	Apple leaf (13,689)	4 apple classes	Improved AlexNet	97.62%
Jiang et al. [5]	Apple leaf (2,637)	5 apple classes	IN-ResNet	94.65%
Zhong & Zhao [20]	PlantVillage (~2,000)	4 apple classes	DenseNet121	93.71%
Bi & Wang [21]	PlantVillage (~2,000)	4 apple classes	ResNet50	97.80%
Yan et al. [22]	PlantVillage (~2,000)	4 apple classes	Improved ResNet50	96.55%
Chen et al. [23]	PlantVillage (~2,000)	Multi-crop classes	VGG19 Transfer	98.24%
Albahli et al. [24]	Custom+PV (~2,000)	Multi-class leaves	DenseNet-based	98.70%
Howard et al. [28]	ImageNet	Mobile tasks	MobileNetV1	High efficiency
Proposed AppleNetV1	Merged Dataset	7 classes (6 apple	Custom CNN	99.33%

(Ours)	(13,470)	+ 1 background)

Table 5: Comparison of proposed AppleNetV1 with existing studies in the literature.

The experimental results present an interesting trade-off between model parameters footprint, classification accuracy, and execution speed. Although ResNet50 achieves the highest accuracy of 99.78%, its larger parameter footprint (96.65 MB) and slower CPU speed (44.0 ms) present deployment challenges on local edge devices. In contrast, MobileNetV2 (12.97 MB) and EfficientNetB0 (20.05 MB) offer highly optimized sizes. EfficientNetB0 is particularly impressive, achieving 99.70% accuracy with a small footprint. The proposed AppleNetV1 baseline is extremely fast on CPU (9.6 ms) due to its simple sequential layers, but its dense classification head leads to a large weights size of 298.61 MB. To validate model transparency, Explainable AI (XAI) using Grad-CAM [30] was utilized. By computing gradients with respect to the last convolutional layer (conv2d_3), we mapped the heatmaps showing where the model focuses. The Grad-CAM heatmaps for Alternaria, Apple Scab, Black Rot, and Cedar Apple Rust show that the model concentrates its focus on leaf spots, lesions, and necrosis, proving that it is learning actual biological signs of diseases instead of background artifacts [31].

7. Conclusions and Future Work

7.1. Conclusions

In this study, we presented AppleNetV1, an efficient and lightweight custom CNN designed for automated apple leaf disease detection. The system successfully classifies seven categories of apple leaves, including six disease classes and a non-apple background class. Through extensive evaluations, the proposed AppleNetV1 achieved a high classification accuracy of 99.33% with an extremely low CPU inference latency of 9.6 ms. Comparisons with state-of-the-art pretrained architectures proved that AppleNetV1 offers a highly optimal balance of size, speed, and accuracy. Furthermore, the integration of a multilingual Streamlit web interface and a Bengali prescription system makes this model highly accessible for practical agricultural deployment.

7.2. Limitations

The primary limitation of this work is the large weight size of AppleNetV1 (298.61 MB), which is caused by the wide dense layers in its classification head. Additionally, the model was trained under supervised laboratory conditions and has not been tested in real-world outdoor lighting and shadows.

7.3. Future Work

Future investigations will focus on compressing AppleNetV1 using pruning and post-training quantization to reduce its memory footprint. We also plan to implement an Out-of-Distribution

(OOD) detector and package the compressed model into a native mobile application for offline field diagnostics.

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